

Name : SHARANKUMAR H N

Reg : 2022BCSE07AED810

Section : Machine Learning B

Engineering Ethics Case Discussion

Case 1: Pressure Vessel Design

An engineer must design a pressure vessel according to a code requiring minimum 20 mm thickness. The customer requests a thinner design to reduce costs.

Answer:

The engineer must not make the vessel thinner than 20 mm. Design codes ensure safety and reliability. A thinner vessel risks catastrophic failure and accidents. The engineer should explain this to the customer and suggest safe cost-saving alternatives like better materials or manufacturing methods.

The engineer's primary duty is public safety. Design codes protect end users. Clear communication with the customer is essential to find solutions that balance cost and safety.

Case 2: Hazardous Waste Disposal

An engineer is responsible for chemical waste disposal. The CEO orders dumping it into a river to cut costs.

Answer:

The engineer must refuse. River dumping is illegal and harms the environment and public health. Legal disposal methods must be used. If needed, report to higher authorities. Protecting the environment and public health is non-negotiable.

Engineers must follow environmental laws regardless of pressure from management. Legal, safe disposal methods exist and must be prioritized over short-term savings.

Conclusion:

Engineers must prioritize safety, legality, and environmental protection above cost pressures. Ethical decisions protect everyone involved and maintain professional integrity.